

Data Structures in Java

Lecture 10: Proofs by Structural Induction

10/10/2016

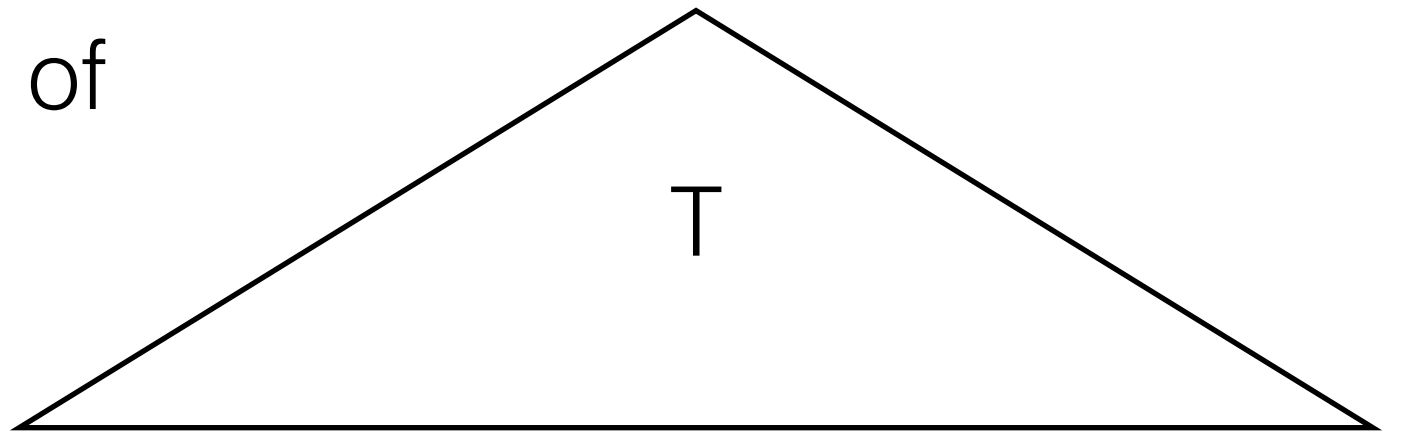
Daniel Bauer

Proofs by Induction

- We are proving a theorem **T**. (“this property holds for all cases.”).
- Step 1: Base case. We know that **T** holds (trivially) for some small value.
- Step 2: Inductive step:
 - Inductive Hypothesis: Assume **T** holds for all cases up to some limit k .
 - Show that **T** also holds for $k+1$.
- This proves that **T** holds for any k .

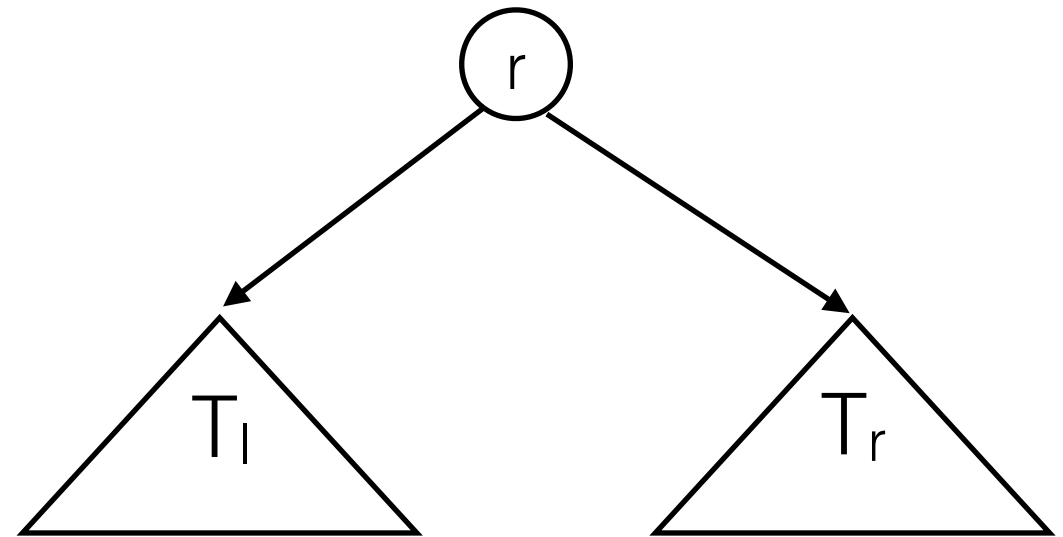
Binary Tree

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 - A root node r .
 - zero, one, or two subtrees.



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Structural Induction over Binary Trees

- We want to show that a property holds for all binary trees.
- Base case: the property holds for a single node.
- Inductive step:
 - Assume the property holds for each possible subtree.
 - Show that it holds for all trees by combining the subtrees with a parent.

Example Proof

- Want to show:
A *perfect* binary tree of height h has $2^{h+1} - 1$ nodes.
- Base case:
A tree of height 0 has $2^{0+1} - 1 = 1$ nodes.
- Inductive step:
 - Assume any perfect binary tree tree of height k has $2^{k+1} - 1$ nodes .
 - Show that any perfect binary tree of height $k + 1$ has $2^{k+2} - 1$ nodes. (next slide)

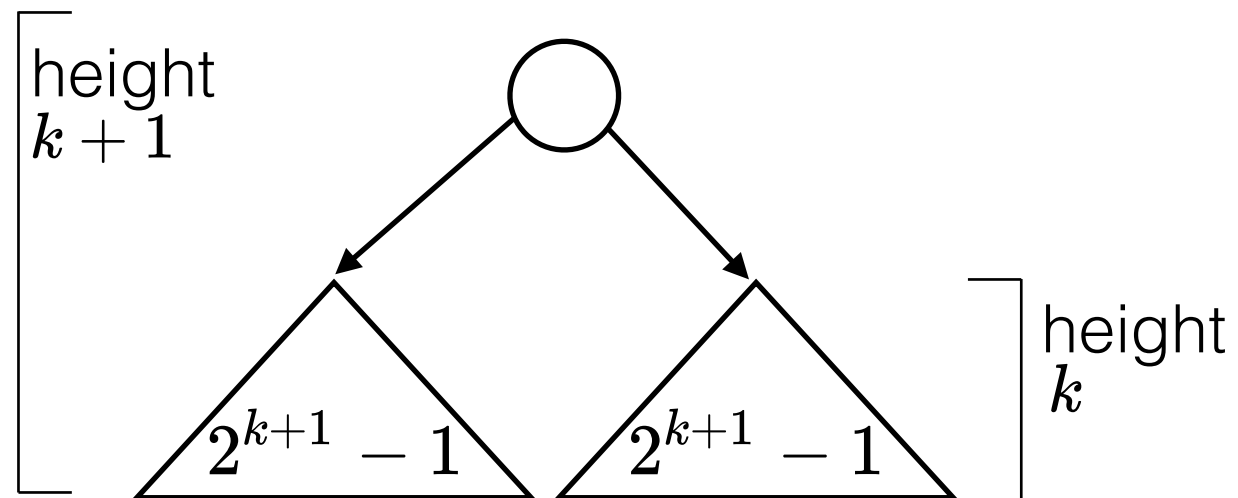
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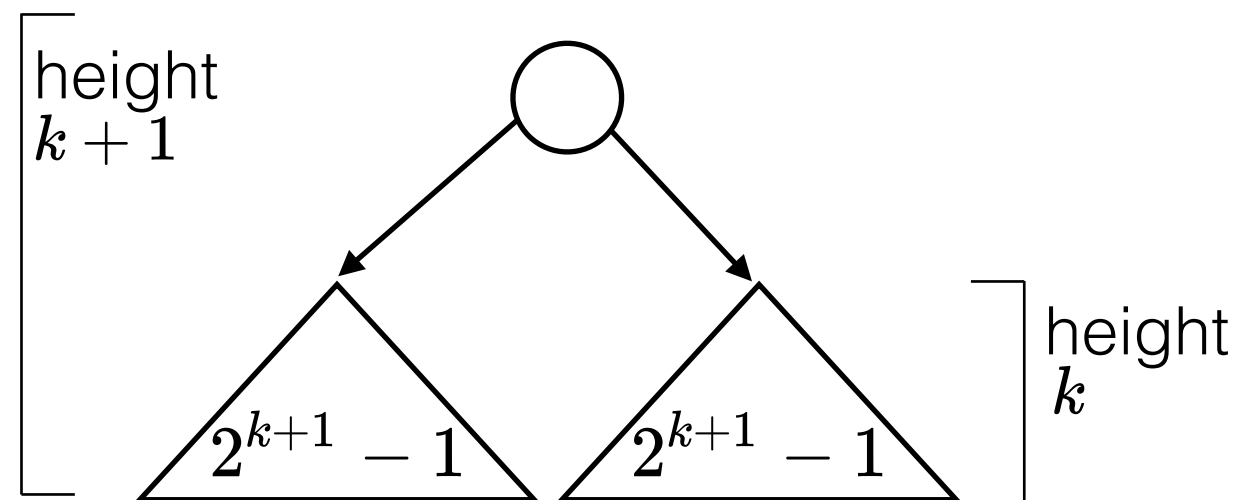
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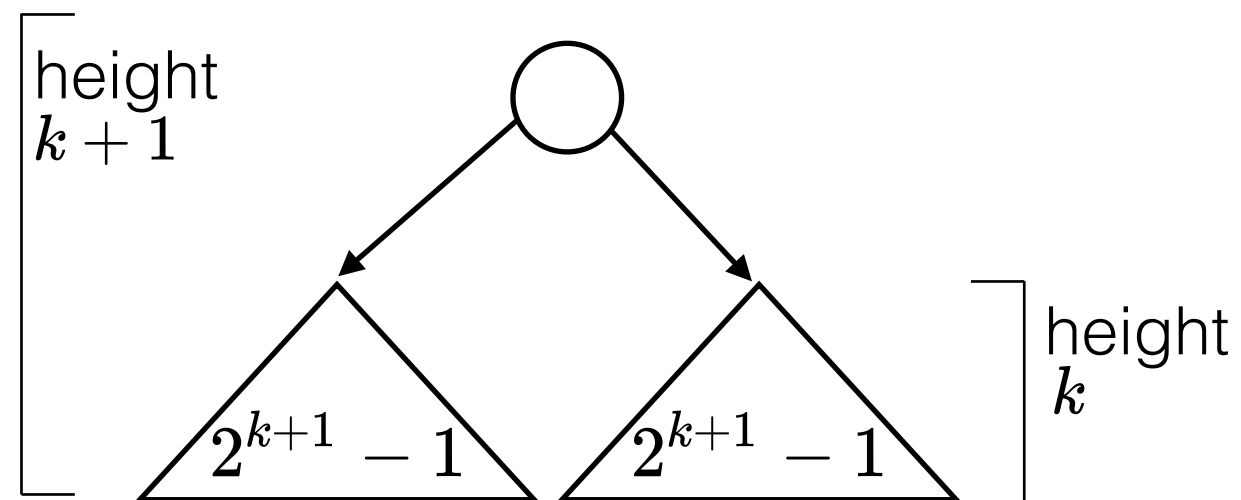


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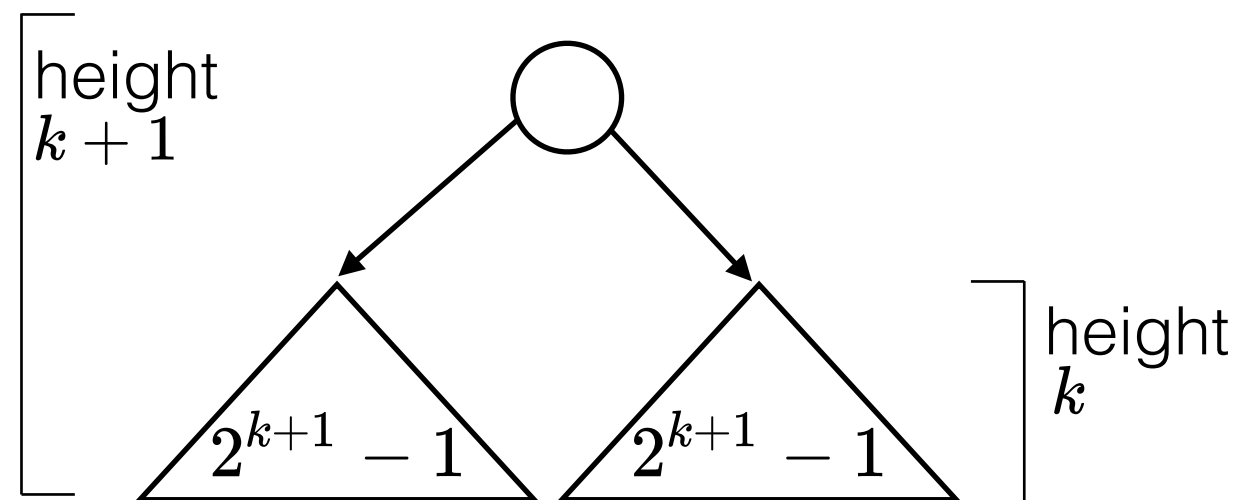


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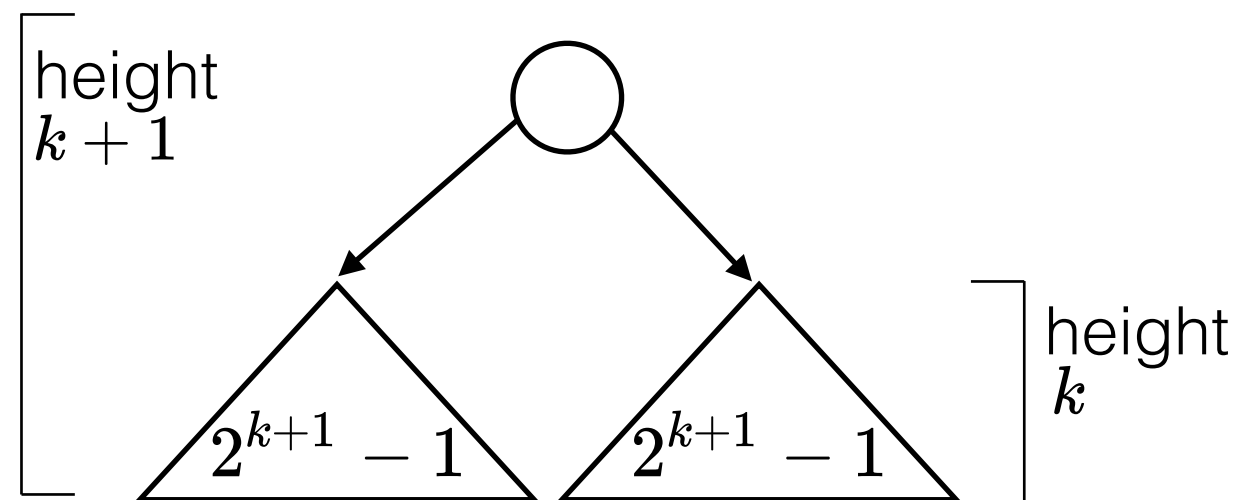
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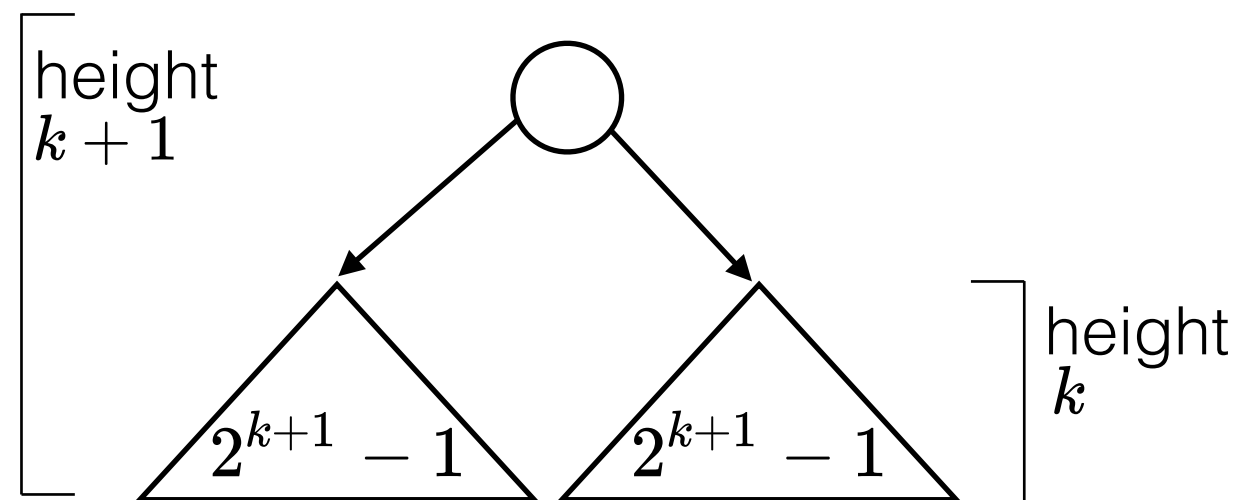
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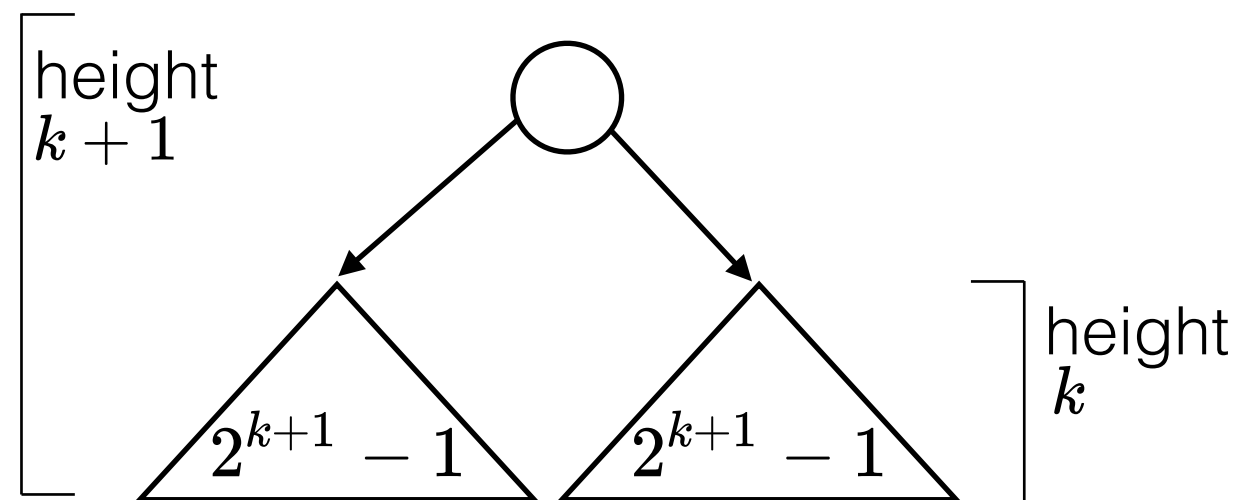
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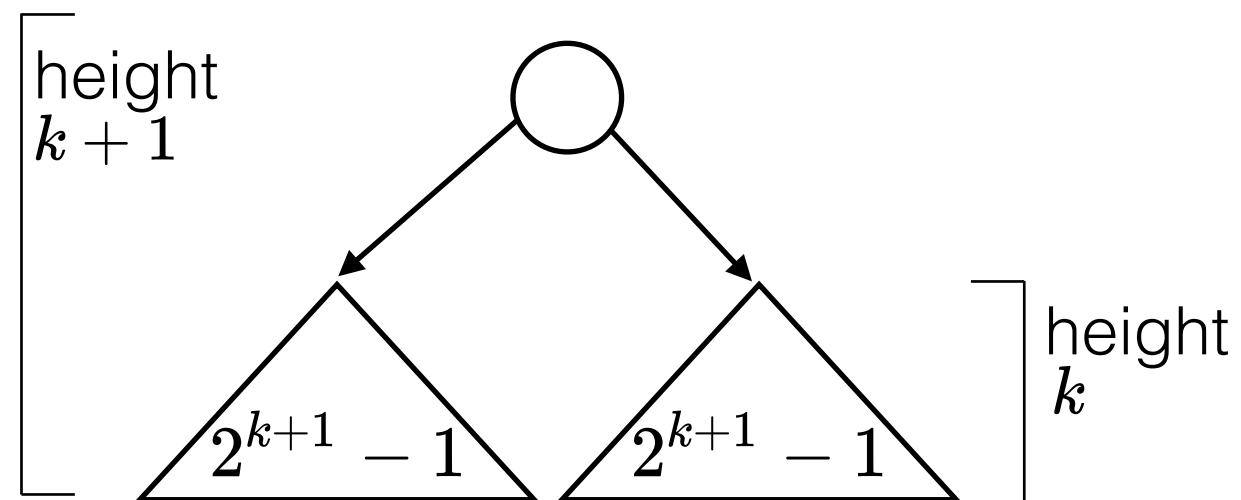
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Minimum Height of a Binary Tree

- A binary tree of height h has *at most* $n = 2^{h+1} - 1$ nodes.
- Therefore, a binary tree of n nodes has at least height $\log(n + 1) - 1$.

$$n + 1 = 2^{h+1}$$

$$\log(n + 1) = h + 1$$

$$\log(n + 1) - 1 = h$$